

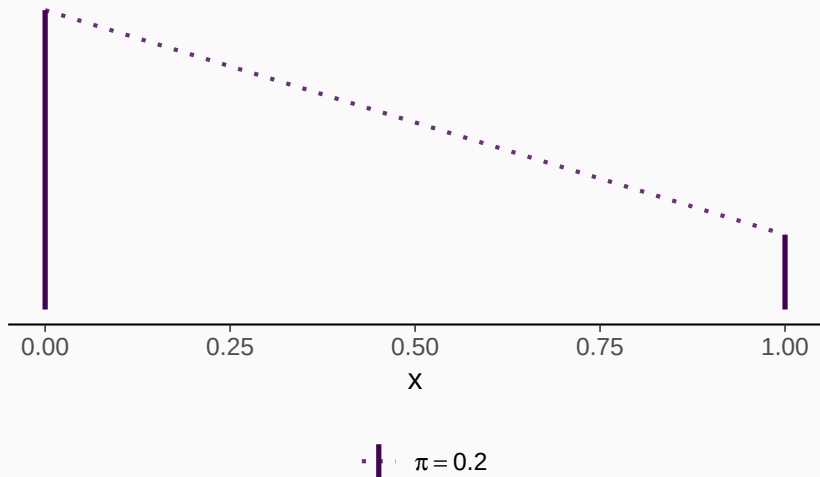
Simulation-Based Inference

Sampling from simple distributions

Paul Bürkner

Sampling from univariate discrete distributions

Bernoulli distribution



Density for $x \in \{0, 1\}$:

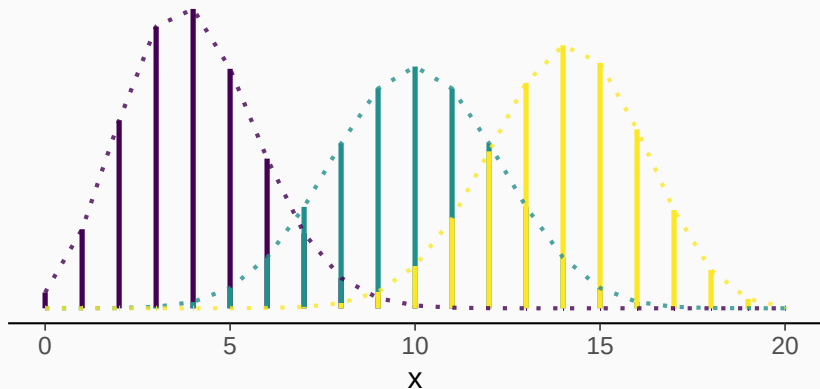
$$p(x \mid \pi) = \pi^x (1 - \pi)^{1-x}$$

Parameters: $\pi \in [0, 1]$ (event probability; mean)

Sampling:

- Divide the integer interval $I = [0, m]$ into two sets I_0 and I_1
- Choose I_1 such that $\pi \approx |I_1|/|I|$
- Set $x = 1$ if the RNG samples a value in I_1

Binomial distribution



■ $N=20, \pi=0.2$ ■ $N=20, \pi=0.5$ ■ $N=20, \pi=0.7$

Density for $x \in \{0, 1, \dots, N\}$:

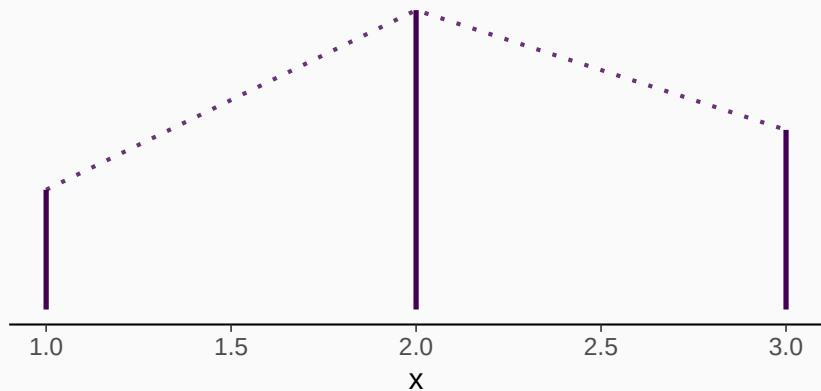
$$p(x \mid N, \pi) = \binom{N}{x} \pi^x (1 - \pi)^{N-x}$$

Parameters: $N \in \mathbb{N}$ (sample size), $\pi \in (0, 1)$ (event probability)

Sampling:

- A binomial variable is the sum of N i.i.d. Bernoulli variables
- Sample N i.i.d. Bernoulli variables with event probability π and sum them up

Categorical distribution



· | $\pi_1 = 0.2, \pi_2 = 0.5, \pi_3 = 0.3$

Density for $x \in \{1, 2, \dots, K\}$:

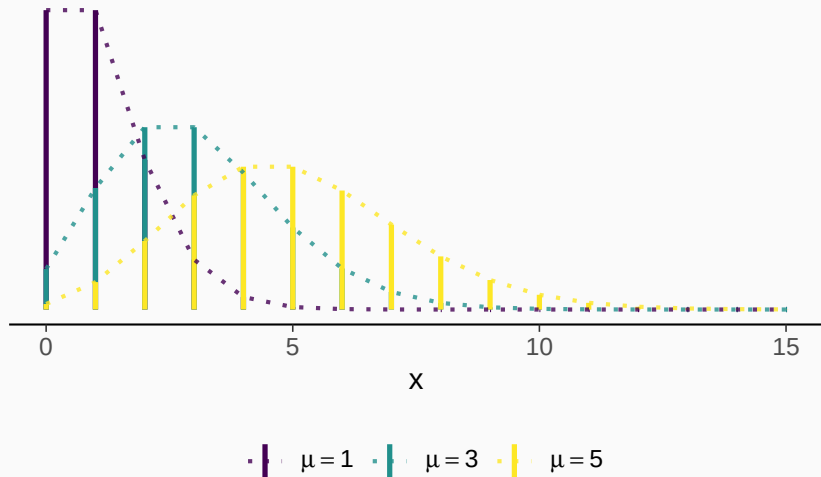
$$p(x \mid \pi) = \pi_x$$

Parameters: $\pi \in [0, 1]^K, \sum_k^K \pi_k = 1$ (event probabilities; means)

Sampling:

- Divide the integer interval $I = [0, m]$ into K sets $I_1, \dots, I_K$
- Choose I_k such that $\pi_k \approx |I_k|/|I|$
- Set $x = k$ if the RNG samples a value in I_k

Poisson distribution



Poisson distribution

Density $x \in \mathbb{N}_0$:

$$p(x \mid \mu) = \frac{\mu^x}{x!} \exp(-\mu)$$

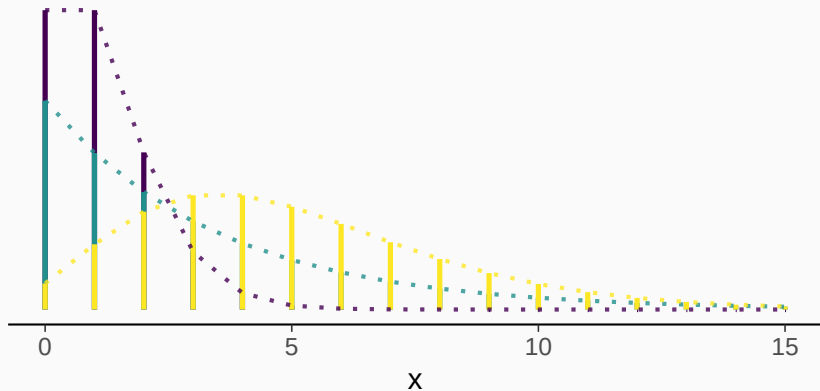
Parameters: $\mu > 0$ (mean)

Sampling:

- Truncate the Poisson such that its support is finite: Set $p(x \mid \mu) = 0$ for all $x > K_{\max}$ with integer $K_{\max}$ large enough
- Sample from categorical with $K_{\max} + 1$ categories with probabilities $\pi_k = p(k \mid \mu)$ for $k < K_{\max}$ and

$$\pi_{K_{\max}} = 1 - \sum_{k=0}^{K_{\max}-1} \pi_k$$

Negative binomial distribution



· | $\pi = 0.95, \phi = 20$ · | $\pi = 0.25, \phi = 1$ · | $\pi = 0.5, \phi = 5$

Negative binomial distribution

Distribution of the number of failures in a sequence of i.i.d. Bernoulli trials with probability π before a specified number of successes ϕ occurs

Density $x \in \mathbb{N}_0$:

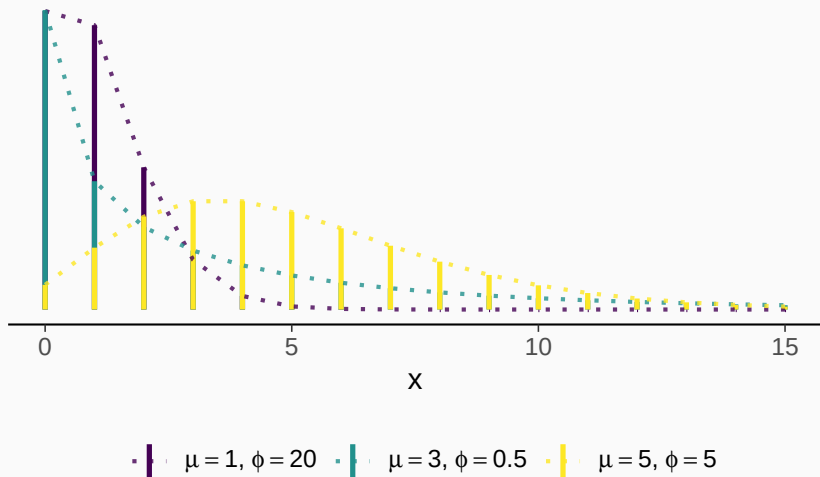
$$p(x \mid \pi, \phi) = \binom{x + \phi - 1}{x} (1 - \pi)^x \pi^\phi$$

Parameters: $\pi \in [0, 1]$ (success probability), $\phi > 0$ (number of successes)

Sampling (for integer ϕ):

- Sample i.i.d. Bernoulli variables with probability π until ϕ successes have occurred
- Count the number of failures in the sequence

Negative-binomial distribution (mean parameterization)



Negative binomial distribution (mean parameterization)

Density $x \in \mathbb{N}_0$:

$$p(x \mid \mu, \phi) = \binom{x + \phi - 1}{x} \left(\frac{\mu}{\mu + \phi} \right)^x \left(\frac{\phi}{\mu + \phi} \right)^\phi$$

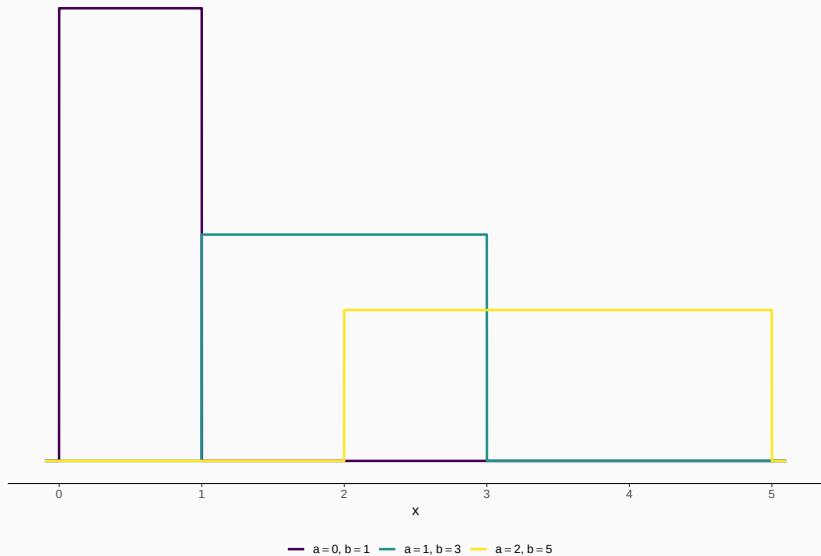
Parameters: $\mu > 0$ (mean), $\phi > 0$ (shape; overdispersion)

Sampling (for integer ϕ):

- Transform μ into parameter $\pi = \phi / (\phi + \mu)$ from the basic negative binomial distribution
- Sample from the basic negative binomial distribution

Sampling from univariate continuous distributions

Continuous uniform distribution



Continuous uniform distribution

A continuous density function of the type

$$p(x \mid a, b) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$$

where $a < b$ are real numbers, defines the density of the continuous uniform distribution on the interval $[a, b]$.

If z is discretely uniform in $[0, m]$, then

$$y := \frac{z}{m},$$

is (approximately) continuously uniform in $[0, 1]$. If y is continuously uniform in $[0, 1]$, then

$$x := (b - a) y + a$$

is continuously uniform in $[a, b]$.

Reminder: Cumulative Distribution function (CDF)

The cumulative distribution function (CDF), often denoted as F , describes the probability that $x \leq \tilde{x}$ for a given $\tilde{x}$:

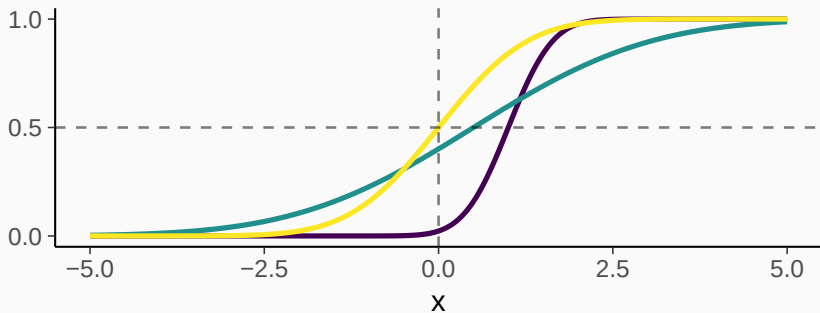
$$F(\tilde{x}) = P(x \leq \tilde{x}) = \int_{-\infty}^{\tilde{x}} p(x) dx$$

The complement of the cumulative distribution function (CCDF) is computed as

$$1 - F(\tilde{x}) = 1 - P(x \leq \tilde{x}) = P(x > \tilde{x}) = \int_{\tilde{x}}^{\infty} p(x) dx$$

CDF example: normal distribution

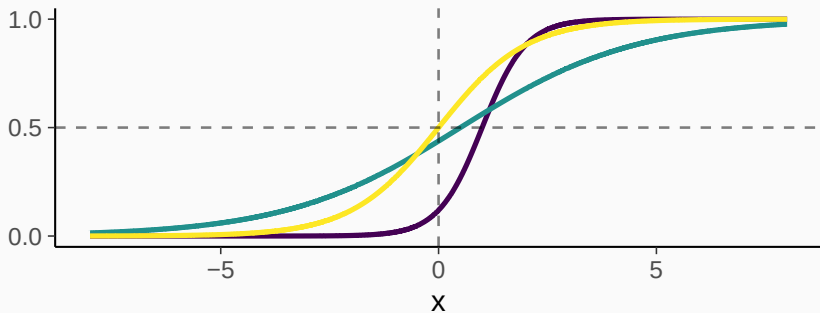
$$F(x) = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2}\sigma} \right) \right) \quad \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt$$



— $\mu = 1, \sigma = 0.5$ — $\mu = 0.5, \sigma = 2$ — $\mu = 0, \sigma = 1$

CDF example: logistic distribution

$$F(x) = \frac{1}{1 + \exp(-(x - \mu)/\sigma)}$$



— $\mu = 1, \sigma = 0.5$ — $\mu = 0.5, \sigma = 2$ — $\mu = 0, \sigma = 1$

For any $\pi \in [0, 1]$, the corresponding quantile $q_\pi \in \Omega_x$ is defined as any value that satisfies

$$P(x \leq q_\pi) \geq \pi \quad \text{and} \quad P(x \geq q_\pi) \geq 1 - \pi$$

In words: (At least) π percent of the x values are smaller than (or equal to) q_π and (at least) $1 - \pi$ percent of the x values are larger than (or equal to) q_π .

Can the CDF help us to compute quantiles?

The quantile function Q_x of a variable x is defined as

$$Q_x : [0, 1] \rightarrow \Omega_x, \quad Q_x(\pi) = q_\pi$$

Related to the CDF via $Q_x = F_x^{-1}$ for univariate x

Quantiles are equivariant under *monotonic* transformations f :

$$Q_{f(x)}(\pi) = f(Q_x(\pi))$$

Do you see why?

Quantile function example: Logistic distribution

CDF of the logistic distribution:

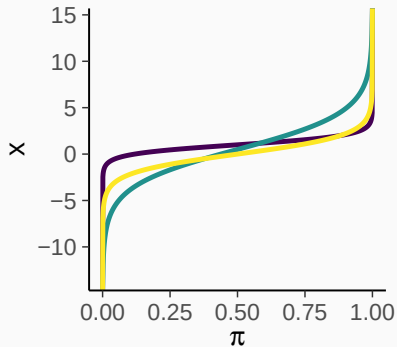
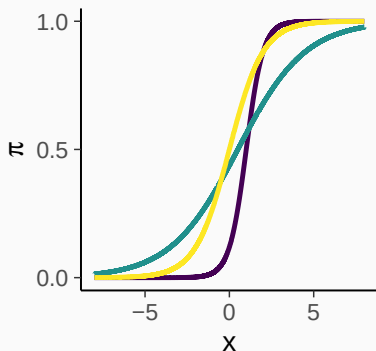
$$\pi = F(x) = \frac{1}{1 + \exp(-(x - \mu)/\sigma)}$$

Quantile function of the logistic distribution:

$$x = Q(\pi) = \mu + \sigma \log\left(\frac{\pi}{1 - \pi}\right)$$

Quantile function example: logistic distribution

CDF (left) and quantile function (right):



■ $\mu = 1, \sigma = 0.5$ ■ $\mu = 0.5, \sigma = 2$ ■ $\mu = 0, \sigma = 10.5$ ■ $\mu = 0.5, \sigma = 2$ ■

NULL

Proposition. Let the distribution function of X be continuous and strictly monotonically increasing. Then $F(X)$ is uniformly distributed on the interval $[0, 1]$.

Proof. The following applies to a random variable Y that is uniformly distributed on the interval $[0, 1]$:

$$P(Y \leq y) = y$$

And the following applies to every distribution function F :

$$P(F(X) \leq F(x)) = P(X \leq x) = F(x),$$

since F is continuous and strictly monotonic.

So the random variable $Y := F(X)$ is equally distributed on $[0, 1]$. □

Quantile functions $Q_x(\pi)$ can be used for sampling from $p(x)$

If

$$\pi \sim \text{uniform}(0, 1)$$

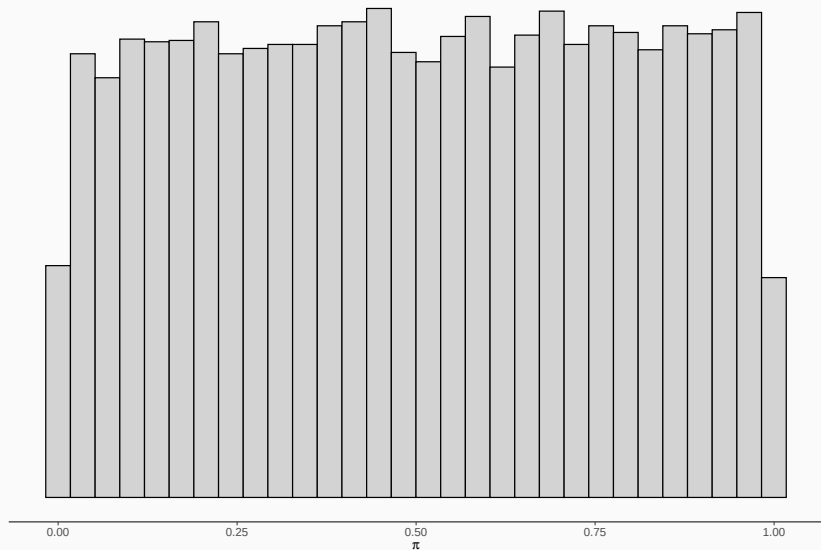
then

$$Q_x(\pi) \sim p(x)$$

Sampling of continuous univariate distributions can be reduced (in theory) to sampling from a standard uniform distribution

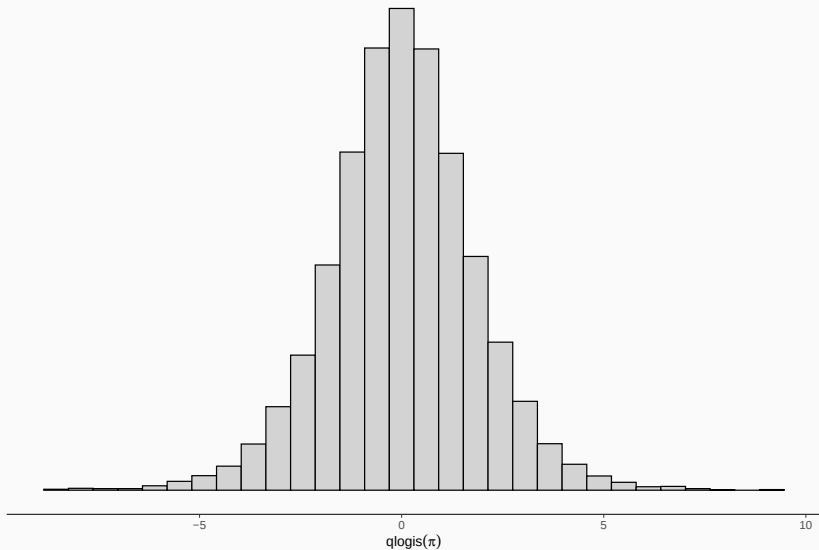
Inverse CDF sampling: Illustration

Standard uniform samples $\pi^{(s)} \sim \text{uniform}(0, 1)$



Inverse CDF sampling: Logistic distribution

Standard logistic samples computed as $x^{(s)} = Q_{\text{logistic}(0,1)}(\pi^{(s)})$



Quantile function example: Normal distribution

CDF of the normal distribution:

$$\pi = F(x) = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2}\sigma} \right) \right)$$

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt$$

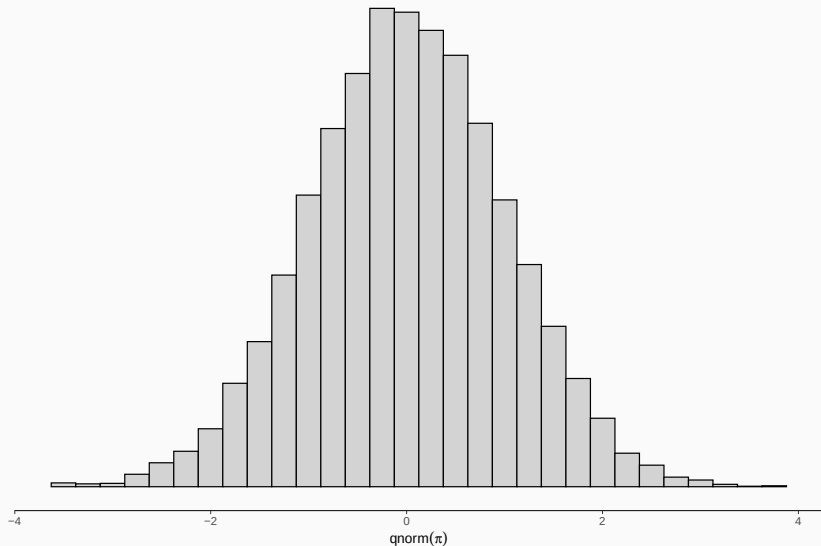
Quantile function of the normal distribution:

$$x = Q(\pi) = \mu + \sqrt{2}\sigma \operatorname{erf}^{-1}(2\pi - 1)$$

where erf^{-1} is the inverse error function

Inverse CDF sampling: Normal distribution

Standard normal samples computed as $x^{(s)} = Q_{\text{normal}(0,1)}(\pi^{(s)})$



Approximate sampling from normal distributions

Central Limit theorem: The sum of a sufficiently well behaved sequence of i.i.d. random variables will converge to a normal distribution.

Example: Sample from a standard normal distribution via standard uniform variables.

Let u_i be uniform in $[0, 1]$, then

$$x := \frac{(\sum_{i=1}^K u_i - K/2)}{\sqrt{K/12}}$$

is approximately standard normal for sufficiently large K

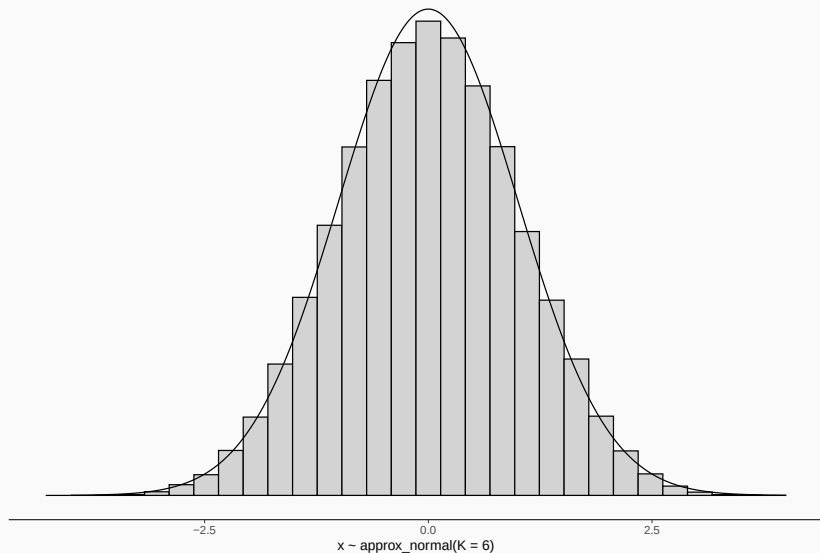
Approximate sampling from normal distributions

Approximate normal distribution for $K = 3$:



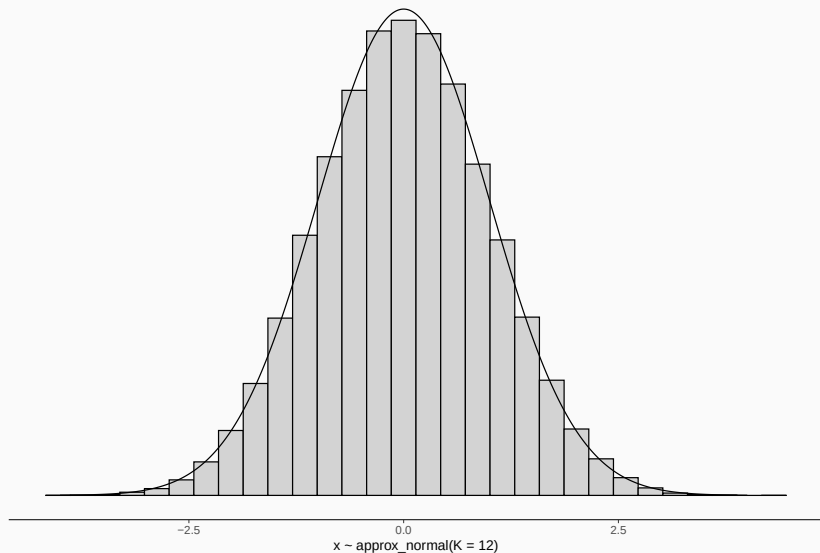
Approximate sampling from normal distributions

Approximate normal distribution for $K = 6$:

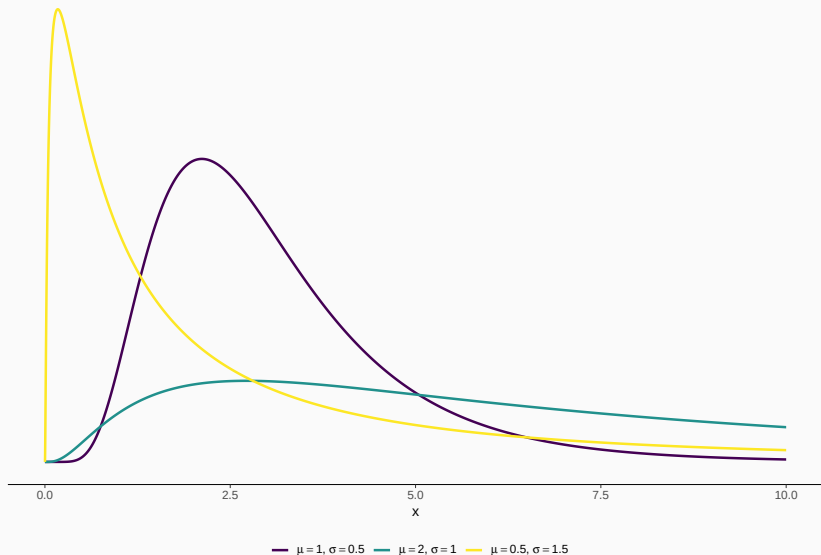


Approximate sampling from normal distributions

Approximate normal distribution for $K = 12$:



Lognormal distribution



Density for $x > 0$:

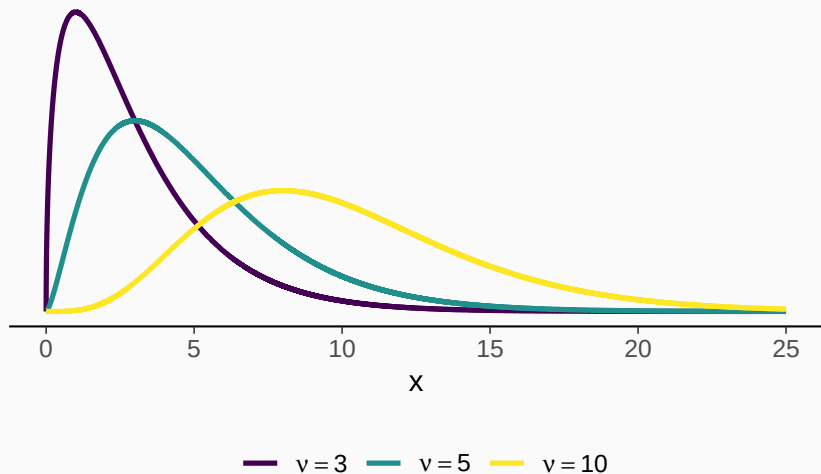
$$p(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma x} \exp\left(-\frac{1}{2} \left(\frac{\log(x) - \mu}{\sigma}\right)^2\right)$$

Parameters: $\mu \in \mathbb{R}$ (log-mean), $\sigma > 0$ (log-standard deviation)

Sampling:

- Sample $z \sim \text{normal}(\mu, \sigma)$
- Compute $x = \exp(z)$

χ^2 distribution



χ^2 distribution

A χ^2 variable is the sum of independent, squared standard normal variables.

Density for $x \in \mathbb{R}_+$:

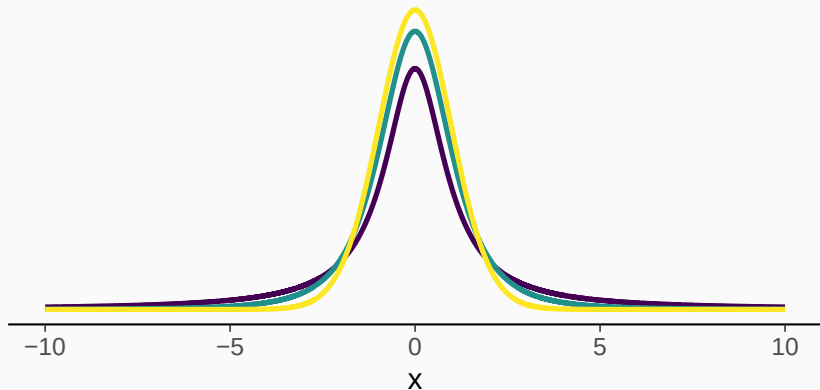
$$p(x \mid \nu) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\nu/2-1} \exp(-x/2)$$

Parameters: $\nu > 0$ (degrees of freedom)

Sampling:

- Sample $z_1, \dots, z_K \sim \text{normal}(0, 1)$
- Compute $x = \sum_{k=1}^K z_k^2$

Student-t distribution



— $v = 1, \mu = 0, \sigma = 1$ — $v = 3, \mu = 0, \sigma = 1$ — $v = 30, \mu = 0, \sigma = 1$

Student-t distribution

Density for $x \in \mathbb{R}$:

$$p(x \mid \nu, \mu, \sigma) = \frac{\Gamma((\nu + 1)/2)}{\Gamma(\nu/2)} \frac{1}{\sqrt{\nu\pi}\sigma} \left(1 + \frac{1}{\nu} \left(\frac{x - \mu}{\sigma}\right)^2\right)^{-(\nu+1)/2}$$

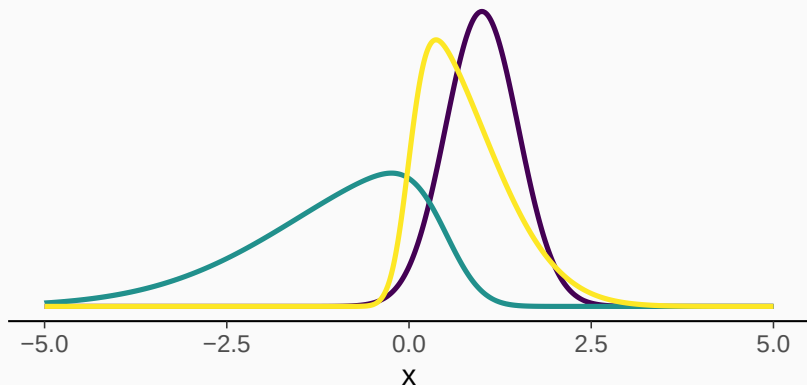
Parameters: $\nu > 0$ (degrees of freedom), $\mu \in \mathbb{R}$ (location), $\sigma > 0$ (scale)

Sampling:

- Sample $u \sim \chi^2(\nu)$ and $z \sim \text{normal}(0, 1)$
- Compute $x = \mu + z\sigma\sqrt{\nu/u}$

Question: Any ideas why sampling from the Student-t works like that?

Skew-normal distribution



— $\xi = 1, \omega = 0.5, \alpha = 0$ — $\xi = 0.5, \omega = 2, \alpha = -5$ — $\xi = 0, \omega = 1, \alpha = 5$

Skew-normal distribution

Density for $x \in \mathbb{R}$:

$$p(x \mid \mu, \sigma, \alpha) = \frac{1}{\sqrt{2\pi}\omega} \exp\left(-\frac{1}{2}\left(\frac{x-\xi}{\omega}\right)^2\right) \left(1 + \operatorname{erf}\left(\alpha \left(\frac{x-\xi}{\omega\sqrt{2}}\right)\right)\right)$$

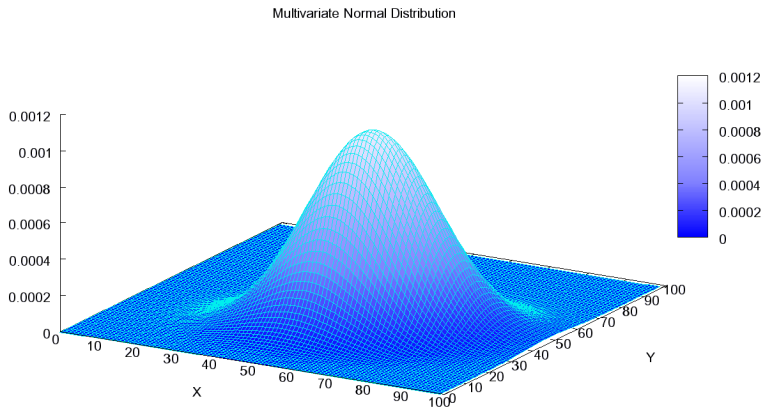
Parameters: $\xi \in \mathbb{R}$ (location), $\omega > 0$ (scale), $\alpha \in \mathbb{R}$ (skewness)

Sampling:

- Sample $z_1, z_2 \sim \text{normal}(0, 1)$
- Obtain indices i for which $z_{2i} > \alpha z_{1i}$
- For these i , set $z_{1i} \leftarrow -z_{1i}$ (sign change)
- Compute $x = \xi + \omega z_1$

Sampling from multivariate continuous distributions

Multivariate normal distribution



Source: https://en.wikipedia.org/wiki/Multivariate_normal_distribution

Multivariate normal distribution

Density for $x \in \mathbb{R}^K$:

$$p(x \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^K |\Sigma|}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

Parameters: $\mu \in \mathbb{R}^K$ (mean vector), $\Sigma \in \mathbb{R}^{K \times K}$, $\Sigma \succ 0$ (covariance matrix)

Sampling:

- Sample $z_1, \dots, z_K \sim \text{normal}(0, 1)$
- Gather them all in a vector $z = (z_1, \dots, z_K)$
- Compute $x = \mu + Lz$

where L is the Cholesky factor of C , that is, $C = LL^T$

Multivariate Student-t distribution

Density for $x \in \mathbb{R}^K$:

$$p(x \mid \nu, \mu, \Sigma) = \frac{1}{\pi^{K/2}} \frac{1}{\nu^{K/2}} \frac{\Gamma((\nu + K)/2)}{\Gamma(\nu/2)} \frac{1}{\sqrt{|\Sigma|}} \left(1 + \frac{1}{\nu} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right)^{-(\nu + K)/2}$$

Parameters: $\nu > 0$ (degrees of freedom), $\mu \in \mathbb{R}^K$ (location vector),
 $\Sigma \in \mathbb{R}^{K \times K}$, $\Sigma \succ 0$ (scale matrix)

Sampling:

- Sample $u \sim \chi^2(\nu)$ and $z \sim \text{MultiNormal}(0, \Sigma)$
- Compute $x = \mu + z\sqrt{\nu/u}$

Importance Sampling

Importance sampling

(Almost) every quantity of interest related to x is an expectation over $p(x)$:

$$\mathbb{E}_{p(x)}(f(x)) = \int f(x) p(x) \, dx$$

Switch the distribution (from p to q) over which to integrate:

$$\mathbb{E}_{p(x)}(f(x)) = \int f(x) r(x) q(x) \, dx$$

with importance ratios

$$r(x) = \frac{p(x)}{q(x)}$$

Self-normalized importance sampling

Problem: Normalizing constant Z_p of $p(x)$ may be unavailable

We usually only have access to the unnormalized density $\tilde{p}(x)$, such that $p(x) = \tilde{p}(x)/Z_p$

We can compute

$$Z_p = \int \tilde{p}(x) dx = \int r(x) q(x) dx$$

with importance ratios

$$r(x) = \frac{\tilde{p}(x)}{q(x)}$$

This results in the self-normalized importance sampling approximation:

$$\mathbb{E}_{p(x)}(f(x)) = \frac{\int f(x) r(x) q(x) dx}{\int r(x) q(x) dx}$$

In the following, we will always use self-normalization

(Self-normalized) importance sampling

- Sample S draws $x^{(s)} \sim q(x)$ from the proposal distributions
- Compute importance ratios $r(x^{(s)}) = p(x^{(s)})/q(x^{(s)})$

Then we can approximate

$$\mathbb{E}_{p(x)}(f(x)) \approx \frac{\sum_{s=1}^S f(x^{(s)}) r(x^{(s)})}{\sum_{s=1}^S r(x^{(s)})} = \sum_{s=1}^S f(x^{(s)}) w(x^{(s)})$$

with

$$w^{(s)} = w(x^{(s)}) = \frac{r(x^{(s)})}{\sum_{s=1}^S r(x^{(s)})}$$

Challenge: The importance ratios $r(x^{(s)})$ may be quite unstable

(Self-normalized) importance resampling

Having obtained the weights $w^{(s)}$, we can approximate any expectation over $p(x)$ with an weighted empirical mean over $q(x)$:

$$\mathbb{E}_{p(x)}(f(x)) \approx \sum_{s=1}^S w^{(s)} f(x^{(s)}) \quad \text{with} \quad x^{(s)} \sim q(x)$$

Instead of using $w^{(s)}$ to compute weighted means directly, we can also perform weighted resampling of the proposal draws $x^{(s)}$ (*importance resampling*):

$$\tilde{x}^{(s)} \sim \text{multinomial}(S, k = (x^{(1)}, \dots, x^{(S)}), p = (w^{(1)}, \dots, w^{(S)}))$$

Then approximately $\tilde{x}^{(s)} \sim p(x)$ and

$$\mathbb{E}_{p(x)}(f(x)) \approx \frac{1}{S} \sum_{s=1}^S f(\tilde{x}^{(s)})$$

Example: Importance sampling from a normal distribution

Target: $\text{normal}(0, 1)$; proposal: $\text{normal}(0.5, 2)$

```
S <- 10000
```

```
x <- rnorm(S, mean = 0.5, sd = 2)
```

```
r <- dnorm(x, mean = 0, sd = 1) /
```

```
  dnorm(x, mean = 0.5, sd = 2)
```

```
w <- r / sum(r)
```

```
(mean_new <- sum(x * w)) # analytic = 0
```

```
[1] -0.001899593
```

```
(var_new <- sum((x - mean_new)^2 * w)) # analytic = 1
```

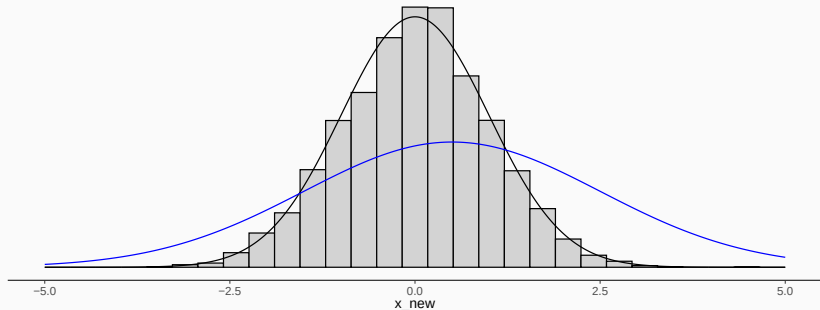
```
[1] 0.9927717
```

Example: Importance sampling from a normal distribution

Perform importance resampling: $\vdots$ {cell}

```
x_new <- sample(x, S, replace = TRUE, prob = w)
```

$\vdots$



Black: target distribution; blue: proposal distribution